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Implement Iris Flower Classification using Naive Bayes classifier

Code :

```
import math

data = [
    ([5.1, 3.5, 1.4, 0.2], "Setosa"),
    ([4.9, 3.0, 1.4, 0.2], "Setosa"),
    ([6.4, 3.2, 4.5, 1.5], "Versicolor"),
    ([6.9, 3.1, 4.9, 1.5], "Versicolor"),
    ([6.3, 3.3, 6.0, 2.5], "Virginica"),
    ([5.8, 2.7, 5.1, 1.9], "Virginica")
]

test = [5.9, 3.0, 4.2, 1.5]

classes = ["Setosa", "Versicolor", "Virginica"]

def mean(x):
    return sum(x) / len(x)

def variance(x):
    m = mean(x)
    return sum((v - m) ** 2 for v in x) / len(x) + 1e-9

def gaussian(x, m, v):
    return math.exp(-(x - m) ** 2 / (2 * v)) / math.sqrt(2 * math.pi * v)

prob = {}

for c in classes:
    samples = [x for x, label in data if label == c]
    p = 1 / 3
    for i in range(4):
        values = [x[i] for x in samples]
        p *= gaussian(test[i], mean(values), variance(values))
    prob[c] = p
```

```
print("Test Data:", test)
```

```
print("Probabilities:", prob)
```

```
print("Predicted Class:", max(prob, key=prob.get))Output :
```

```
Test Data: [5.9, 3.0, 4.2, 1.5]
```

```
Probabilities: {'Setosa': 0.0, 'Versicolor': 0.5791074579694716,  
               'Virginica': 0.0005086039472729686}
```

```
Predicted Class: Versicolor
```

```
=== Code Execution Successful ===
```